

## 5 Wavepacket dynamics

### 5.1 Basis functions for a free particle

A free particle has a continuum of states which are eigenfunctions of the Hamiltonian,

$$\hat{H}_0 = \frac{\hat{p}^2}{2m}. \quad (5.1)$$

The eigenstates of this are also the eigenstates of  $\hat{p}$

$$\hat{p} |p\rangle = p |p\rangle \quad (5.2)$$

$$\hat{H}_0 |p\rangle = E_p |p\rangle, \quad (5.3)$$

where  $E_p = p^2/2m$ .

The position-representation of the eigenstate is

$$\langle x|p\rangle = \phi_p(x) \quad (5.4)$$

and is found by solving

$$\hat{p}\phi_p(x) = -i\hbar \frac{\partial}{\partial x} \phi_p(x) = p\phi_p(x) \quad (5.5)$$

to give

$$\boxed{\langle x|p\rangle = \phi_p(x) = (2\pi\hbar)^{-\frac{1}{2}} e^{+ipx/\hbar}.} \quad (5.6)$$

The normalization term has been chosen such that

$$\langle p'|p\rangle = \int_{-\infty}^{\infty} \phi_{p'}(x)^* \phi_p(x) dx \quad (5.7)$$

$$= (2\pi\hbar)^{-1} \int_{-\infty}^{\infty} e^{-i(p'-p)x/\hbar} dx \quad (5.8)$$

$$= \delta(p' - p) \quad (5.9)$$

This is a subtly different normalization scheme from what is used for discrete states where  $\langle n|n\rangle = 1$ . The momentum eigenstates form a complete basis set such that any one-dimensional wave function (not just those describing free particles) can be written as

$$|\psi\rangle = \int_{-\infty}^{\infty} |p\rangle \langle p|\psi\rangle dp. \quad (5.10)$$

In the same way there are position states

$$\boxed{\langle p|x\rangle = \langle x|p\rangle^* = (2\pi\hbar)^{-\frac{1}{2}} e^{-ipx/\hbar}} \quad (5.11)$$

which are complete such that

$$|\psi\rangle = \int_{-\infty}^{\infty} |x\rangle \langle x|\psi\rangle dx \quad (5.12)$$

and normalized such that

$$\langle x'|x\rangle = \delta(x' - x). \quad (5.13)$$

They are thus like a set of infinitely small pixels which together with the weights  $\langle x|\psi\rangle = \psi(x)$  can define a state  $|\psi\rangle$ .

So the evolution of a free-particle wavepacket is

$$|\psi(t)\rangle = \hat{U}_0(t, t_0) |\psi(t_0)\rangle \quad (5.14)$$

$$= e^{-i\hat{H}_0(t-t_0)/\hbar} |\psi(t_0)\rangle \quad (5.15)$$

$$= e^{-\frac{i\hat{p}^2(t-t_0)}{2m\hbar}} |\psi(t_0)\rangle \quad (5.16)$$

$$= \int e^{-\frac{i\hat{p}^2(t-t_0)}{2m\hbar}} |p\rangle \langle p|\psi(t_0)\rangle dp \quad (5.17)$$

$$= \int |p\rangle e^{-\frac{ip^2(t-t_0)}{2m\hbar}} \langle p|\psi(t_0)\rangle dp. \quad (5.18)$$

We use the word “wavepacket” to emphasize that the wavefunction is a linear combination of eigenstates and therefore has interesting dynamics. In many cases the wavepacket is localized in space at all times.

If we know the position representation of the wavepacket, it is possible to compute the momentum-representation using a Fourier transform:

$$\langle p|\psi(t_0)\rangle = \int \langle p|x\rangle \langle x|\psi(t_0)\rangle dx \quad (5.19)$$

$$= (2\pi\hbar)^{-\frac{1}{2}} \int e^{-ipx/\hbar} \langle x|\psi(t_0)\rangle dx \quad (5.20)$$

Likewise we can convert back using an inverse Fourier transform

$$\langle x|\psi(t_0)\rangle = \int \langle x|p\rangle \langle p|\psi(t_0)\rangle dp \quad (5.21)$$

$$= (2\pi\hbar)^{-\frac{1}{2}} \int e^{ipx/\hbar} \langle p|\psi(t_0)\rangle dp. \quad (5.22)$$

For example, if the initial wavepacket is a Gaussian  $\langle x|\psi(t_0)\rangle = \mathcal{N}^{-1} e^{-\alpha(x-x_0)^2/2 + ip_0(x-x_0)/\hbar}$ , where  $x_0$  is the centre of the wavepacket and  $p_0$  is the expectation value of the momentum, then the momentum representation is

$$\langle p|\psi(t_0)\rangle = \frac{\mathcal{N}^{-1}}{(2\pi\hbar)^{\frac{1}{2}}} \int e^{-ipx/\hbar} e^{-\alpha(x-x_0)^2/2 + ip_0(x-x_0)/\hbar} dx \quad (5.23)$$

$$= \frac{\mathcal{N}^{-1}}{(2\pi\hbar)^{\frac{1}{2}}} \int e^{-ipx_0/\hbar} e^{-\alpha(x-x_0)^2/2 - i(p-p_0)(x-x_0)/\hbar} dx \quad (5.24)$$

$$= \frac{\mathcal{N}^{-1}}{(2\pi\hbar)^{\frac{1}{2}}} \sqrt{\frac{2\pi}{\alpha}} e^{-ipx_0/\hbar} e^{-\frac{(p-p_0)^2}{2\alpha\hbar^2}} \quad (5.25)$$

$$= \frac{\mathcal{N}^{-1}}{(\hbar\alpha)^{\frac{1}{2}}} e^{-ipx_0/\hbar} e^{-\frac{(p-p_0)^2}{2\alpha\hbar^2}} \quad (5.26)$$

which is another Gaussian (with an unimportant phase factor) centred around  $p_0$ , hence confirming that this parameter can be interpreted as the initial momentum.

Therefore the wavepacket at time  $t$  is

$$\langle x|\psi(t)\rangle = \int \langle x|p\rangle e^{-\frac{ip^2(t-t_0)}{2m\hbar}} \langle p|\psi(t_0)\rangle dp \quad (5.27)$$

$$= \frac{\mathcal{N}^{-1}}{\sqrt{2\pi\alpha\hbar^2}} \int e^{ipx/\hbar} e^{-\frac{ip^2(t-t_0)}{2m\hbar}} e^{-ipx_0/\hbar} e^{-\frac{(p-p_0)^2}{2\alpha\hbar^2}} dp \quad (5.28)$$

$$= \frac{\mathcal{N}^{-1}}{\sqrt{2\pi\alpha\hbar^2}} \int e^{ip(x-x_0)/\hbar} e^{-\frac{ip^2(t-t_0)}{2m\hbar}} e^{-\frac{(p-p_0)^2}{2\alpha\hbar^2}} dp \quad (5.29)$$

$$= \mathcal{N}_t^{-1} e^{-\alpha_t(x-x_t)^2/2 + ip_0(x-x_t)/\hbar + ic_t} \quad (5.30)$$

where

$$x_t = x_0 + \frac{p_0}{m}t \quad (5.31)$$

$$\alpha_t = \alpha/(1 + i\hbar\alpha t/m) \quad (5.32)$$

$$c_t = \frac{p_0^2}{2m}t + \frac{i}{2}\hbar \log(1 + i\hbar\alpha t/m) \quad (5.33)$$

So it remains a Gaussian centred around  $x_t$  (which moves like a classical particle), with an ever increasing width.\*

## 5.2 Wavepacket dynamics

In previous sections we have propagated wavefunctions by expanding them as a linear combination of eigenstates. In many cases, one does not want to do this, as solving for the eigenstates can be a difficult problem. It will not be possible to solve the problem analytically in general. Nonetheless, it is possible to evolve the wavefunction numerically using the propagator in an appropriate basis.

$$|\psi(t)\rangle = \hat{U}(t, t_0) |\psi(t_0)\rangle \quad (5.34)$$

First use the property of the propagator:

$$\hat{U}(t, t_0) = \hat{U}(t, t_N) \hat{U}(t_N, t_{N-1}) \cdots \hat{U}(t_2, t_1) \hat{U}(t_1, t_0). \quad (5.35)$$

This is just like saying “travelling forward by a hour is the same as travelling forward by a minute 60 times.”

The Hamiltonian is a sum of kinetic and potential operators, which do not commute with each other.

$$\hat{U}(t_{n+1}, t_n) = e^{-i\hat{H}\epsilon/\hbar} \quad (5.36a)$$

$$= e^{-i(\hat{T}+\hat{V})\epsilon/\hbar} \quad (5.36b)$$

$$\approx e^{-i\hat{V}\epsilon/2\hbar} e^{-i\hat{T}\epsilon/\hbar} e^{-i\hat{V}\epsilon/2\hbar} + \mathcal{O}(\epsilon^3) \quad (5.36c)$$

$$\approx e^{-i\hat{T}\epsilon/\hbar} e^{-i\hat{V}\epsilon/\hbar} + \mathcal{O}(\epsilon^2) \quad (5.36d)$$

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\*You can derive these formulas for yourself using mathematica. Also see the discussion in Tannor, chapter 3.

where we have given two alternative approximations, one more accurate than the other.

Exercise: take Taylor expansions (in  $\epsilon$ ) of all three expressions [equation (5.36b)–(5.36c)] and confirm that the errors are of order  $\epsilon^2$  and  $\epsilon^3$ .\*

It is easy to apply the potential-energy operator in the position representation because  $\hat{V}|x\rangle = V(x)|x\rangle$  and  $\langle x|\hat{V} = \langle x|V(x)$  and likewise it is easy to apply the kinetic-energy operator in the momentum representation because  $\hat{T}|p\rangle = \frac{p^2}{2m}|p\rangle$  and  $\langle p|\hat{T} = \langle p|\frac{p^2}{2m}$ . Therefore in order to propagate by the full Hamiltonian, we alternate between  $\hat{V}$  and  $\hat{T}$  steps and use Fourier transforms [equation (5.19)] to change representation each time. This is called the *split operator* approach.

First we consider the potential-energy term

$$\langle x|\psi^a(t_n)\rangle = \langle x|e^{-i\hat{V}\epsilon/2\hbar}|\psi(t_n)\rangle \quad (5.37)$$

$$= e^{-iV(x)\epsilon/2\hbar} \langle x|\psi(t_n)\rangle \quad (5.38)$$

which simply multiplies the wavefunction by the phase term. Next, we carry out a Fourier transform [equation (5.19)] to obtain  $\langle p|\psi^a(t_n)\rangle$ . Then we apply the kinetic-energy term

$$\langle p|\psi^b(t_n)\rangle = \langle p|e^{-i\hat{T}\epsilon/\hbar}|\psi^a(t_n)\rangle \quad (5.39)$$

$$= e^{-ip^2\epsilon/2m\hbar} \langle p|\psi^a(t_n)\rangle \quad (5.40)$$

This is inverse Fourier transformed [equation (5.21)] to obtain  $\langle x|\psi^b(t_n)\rangle$ . Finally, operation by the second potential-energy term gives

$$\langle x|\psi(t_{n+1})\rangle = \langle x|e^{-i\hat{V}\epsilon/2\hbar}|\psi^b(t_n)\rangle \quad (5.41)$$

$$= e^{-iV(x)\epsilon/2\hbar} \langle x|\psi^b(t_n)\rangle \quad (5.42)$$

You should convince yourself that these steps are equivalent to

$$\langle x|\psi(t_{n+1})\rangle = \langle x|\hat{U}(t_{n+1}, t_n)|\psi(t_n)\rangle \quad (5.43)$$

$$\approx \langle x|e^{-i\hat{V}\epsilon/2\hbar}e^{-i\hat{T}\epsilon/\hbar}e^{-i\hat{V}\epsilon/2\hbar}|\psi(t_n)\rangle \quad (5.44)$$

$$= \iint \langle x|e^{-i\hat{V}\epsilon/2\hbar}e^{-i\hat{T}\epsilon/\hbar}|p\rangle \langle p|e^{-i\hat{V}\epsilon/2\hbar}|x'\rangle \langle x'|\psi(t_n)\rangle dx' dp \quad (5.45)$$

$$= \iint e^{-iV(x)\epsilon/2\hbar} \langle x|p\rangle e^{-ip^2\epsilon/2m\hbar} \langle p|x'\rangle e^{-iV(x')\epsilon/2\hbar} \langle x'|\psi(t_n)\rangle dx' dp \quad (5.46)$$

Ultimately, all of these formulas can be easily derived simply from the completeness relations.

### 5.3 Electronic absorption spectrum

If we have two uncoupled electronic states, the interaction with the light field may cause transitions between the two. For this, we can use the Born–Oppenheimer expansion in terms of the ground and excited states,  $|\phi_g\rangle$  and  $|\phi_e\rangle$ .

The dipole operator of interest is

$$\hat{\mu}_{eg}(\vec{R}) = \langle \phi_e|\vec{\mu}|\phi_g\rangle_{\vec{r}} \quad (5.47)$$

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\*Solutions are given in Tannor, Exercises 11.1 and 11.2.

and when applied to the initial wavefunction  $|\phi_g\rangle |\chi_g(t_0)\rangle$ , leads to the perturbed wavefunction on the electronically excited state:

$$|\chi_e^{(1)}(t)\rangle = \frac{-1}{i\hbar} \int_{t_0}^t e^{-i\hat{H}_e(t-t')/\hbar} \hat{\mu}_{eg} \cdot \vec{E}_0 \cos(\omega t') e^{-i\hat{H}_g(t'-t_0)/\hbar} |\chi_g(t_0)\rangle dt' \quad (5.48)$$

Let us assume that the initial wavefunction  $|\chi_g(t_0)\rangle$  is an eigenstate of  $\hat{H}_g$  with eigenvalue  $\hbar\omega_g$ . Then,

$$|\chi_e^{(1)}(t)\rangle = \frac{-\vec{E}_0}{i\hbar} \cdot \int_{t_0}^t e^{-i\hat{H}_e(t-t')/\hbar} \hat{\mu}_{eg} |\chi_g(t_0)\rangle \cos(\omega t') e^{-i\omega_g(t'-t_0)} dt' \quad (5.49)$$

We can interpret this as taking the initial wavepacket  $\chi_g(t_0)$  and multiplying it by the  $\vec{R}$ -dependent function  $\mu_{eg}$ . This modified wavepacket is then placed in the excited electronic state, and because it is typically no longer at the equilibrium position, starts to move (see Figure 4).

Following a similar approach to that of the last lecture, we can show that the absorption spectrum can be obtained as the Fourier transform of the dipole-dipole correlation function

$$I(\omega) \propto \langle \chi_e^{(1)}(t) | \chi_e^{(1)}(t) \rangle \quad (5.50)$$

$$\propto \iint_{t_0}^t e^{+i\omega_g(t''-t_0)} \cos(\omega t'') \langle \chi_g(t_0) | \hat{\mu}_{eg} e^{+i\hat{H}_e(t-t'')/\hbar} e^{-i\hat{H}_e(t-t')/\hbar} \hat{\mu}_{ge} | \chi_g(t_0) \rangle \cos(\omega t') e^{-i\omega_g(t'-t_0)} dt' dt'' \quad (5.51)$$

$$= \frac{1}{2} \iint_{t_0}^t \langle \chi_g(t_0) | \hat{\mu}_{ge} e^{-i\hat{H}_e(t''-t')/\hbar} \hat{\mu}_{eg} | \chi_g(t_0) \rangle e^{i\omega_g(t''-t')} [\cos \omega(t' + t'') + \cos \omega(t'' - t')] dt' dt'' \quad (5.52)$$

where  $\hat{\mu}_{ge} = \hat{\mu}_{eg}^\dagger = \langle \phi_g | \vec{\mu} | \phi_e \rangle_{\vec{r}}$ . Neglecting the term with  $\cos \omega(t' + t'')$ , taking the limits  $t_0 \rightarrow -\infty$  and  $t \rightarrow \infty$ , and renaming  $t' - t'' = t$ , we obtain

$$I(\omega) \propto \int_{-\infty}^{\infty} \langle \chi_g(-\infty) | \hat{\mu}_{ge} e^{-i\hat{H}_e t/\hbar} \hat{\mu}_{eg} | \chi_g(-\infty) \rangle e^{i\omega_g t} \cos(\omega t) dt \quad (5.53)$$

For an equilibrium ensemble, we write

$$I(\omega) \propto \int_{-\infty}^{\infty} C_{\mu\mu}(t) \cos(\omega t) dt \quad (5.54)$$

where

$$C_{\mu\mu}(t) = \text{Tr}[\hat{\rho} \hat{\mu}_{ge} e^{-i\hat{H}_e(t''-t')/\hbar} \hat{\mu}_{eg} e^{i\hat{H}_g(t''-t')/\hbar}]. \quad (5.55)$$

This is the same as above if  $\hat{\rho} = |\chi_g\rangle\langle\chi_g|$  but also generalizes the result for an equilibrium ensemble.

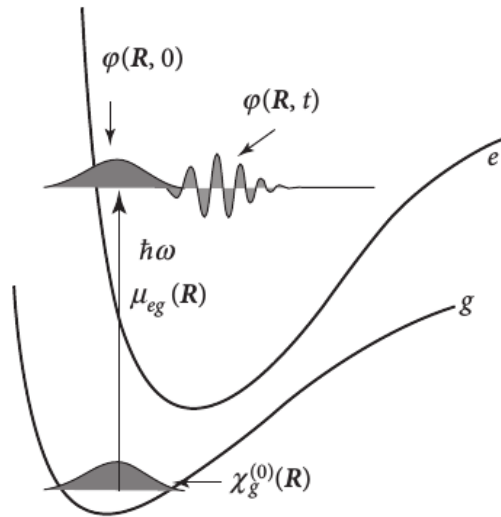


Figure 4: The initial wavepacket on the ground state is  $\chi_g^{(0)}(\mathbf{R})$  and this is modified by the dipole operator to give  $\phi(\mathbf{R}, 0) = \hat{\mu}\chi_g^{(0)}(\mathbf{R})$  which is now placed on the excited state. This evolves to  $\phi(\mathbf{R}, t) = e^{-i\hat{H}_e t/\hbar}\phi(\mathbf{R}, 0)$ . From Heller, *The Semiclassical Way to Dynamics and Spectroscopy*.

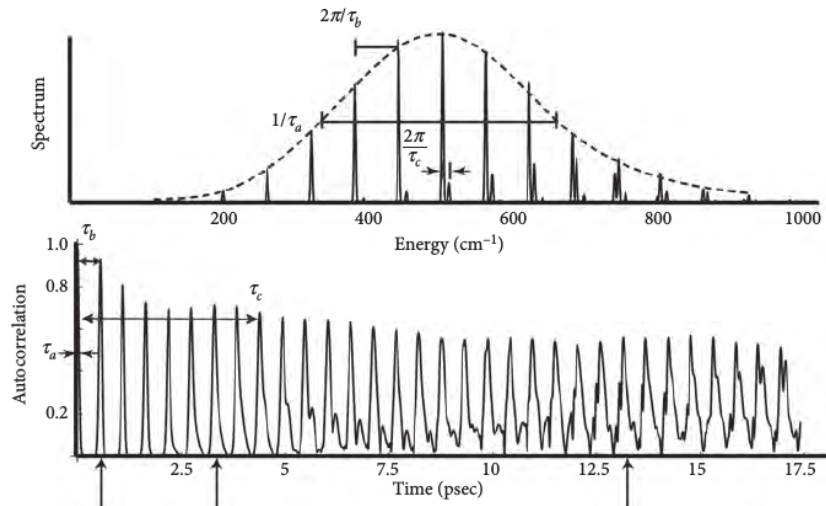


Figure 5: The spectrum and correlation function corresponding to electronic absorption. From Heller, *The Semiclassical Way to Dynamics and Spectroscopy*.

## 5.4 Further Reading

The following textbooks will give you further information on the topics of this lecture:

*Bras and kets*

- Feynman Lectures Volume III chapter 5, <https://www.feynmanlectures.caltech.edu>

*Wavepacket propagation*

- Tannor, Introduction to quantum mechanics: a time-dependent perspective, chapters 2, 3 and 11.7.1

*Electronic absorption spectroscopy*

- Heller, The Semiclassical Way to Dynamics and Spectroscopy, chapter 17
- Tannor, Introduction to quantum mechanics: a time-dependent perspective, section 14.1

Hopefully you may be able to read them on the internet.